about that, we can always appeal again to the fact that particle collisions in the cosmos are and have always been reaching much higher energies than the LHC or any other machine possibly could, so as long as we haven't blinked out of existence yet, our modern equivalent of banging rocks together is really no threat at all.

The difficulty of creating a high enough energy event to directly trigger vacuum decay comes down to the height of the *potential barrier* between our false vacuum and the true one. Going back to the picture of the pebble stuck in a divot, the potential barrier is the bit of land that sticks up to make a divot pocket-shaped. In our current best guess at the true shape of the Higgs potential, the divot is a substantial one, separated from the deeper true-vacuum valley by a very high ridge. The amount of energy it would take to kick the pebble over that ridge (or push the Higgs field over its potential barrier) is so high, it's hardly worth worrying about.

Except … we're living in a universe that doesn't follow those kinds of rules. Our cosmos is fundamentally based on quantum mechanics, and in quantum mechanics, if you're living on a subatomic scale, the path you take to get from one place to another might, very rarely, send you sailing right through solid objects without missing a beat. If you're standing in front of a wall, you might not need to get enough energy to jump over it. You might be able to step right through it instead. Especially if "you" are the Higgs field.

TUNNELING INTO THE ABYSS

Quantum tunneling sounds like science fiction or some obscure theoretical thing that physicists sit around chuckling over while writing down incomprehensible equations. Like, sure, quantum mechanics says that you can't ever really say for sure exactly where a particle is, or what path it's taking as it travels. That means that to get the math to work out, you have to write down and calculate things about *all* the paths, even the outlandish ones that send the particle from one side of the lab to the other by way of a coffee shop three cities over. But that doesn't mean that the particle *really* does that, right?

As it happens, the question of what the particle *really* does is surprisingly hard to answer, and has spurred a decades-long debate about interpretations of quantum mechanics. Where the particle goes on the journey between